Hyperspectral and SAR Image Classification via Graph Convolutional Fusion Network

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***Abstract*— Hyperspectral and synthetic aperture radar (SAR) image classification, aiming to merge multisource information to boost the precision and reliability of land cover classification, has gained increasing attention. Nevertheless, current techniques still exhibit certain limitations in extracting discriminative features and integrating heterogeneous features. In this work, a graph convolutional fusion network (GCFNet) is proposed for hyperspectral and SAR image classification. First, a spectral residual neural network is employed to extract the spectrum information. Then, a dual-branch graph convolutional network (GCN) is developed to extract the spatial information from hyperspectral and SAR images. Finally, a cross-contextual transformer fusion module is created to merge the spectral and spatial information followed by a dense layer to yield the final prediction outcome. To confirm the performance of the GCFNet, experiments on three datasets (e.g., Berlin, Augsburg, and Yellow River) demonstrate that the GCFNet significantly surpasses other representative methods.**

***Index Terms*— Feature fusion, graph convolutional network (GCN), hyperspectral image (HSI), synthetic aperture radar (SAR).**

I. Introduction

INTELLIGENT interpretation of multimodal remote sensing images has attracted much more attention in the remote sensing community [1], [2], [3], [4], [5], [6]. According to different data acquisition methods, remote sensing can be categorized into two primary categories: passive remote sensing and active remote sensing [7], [8], [9], [10]. Passive remote sensing technologies include visible remote sensing, infrared remote sensing, and hyperspectral remote sensing. In particular, hyperspectral image (HSI) encompasses plentiful spectral information, which has been widely applied for ground object classification, environmental monitoring, resource surveys, and other applications [11], [12], [13], [14], [15], [16], [17]. Active remote sensing mainly includes synthetic aperture radar (SAR) and light detection and ranging (LiDAR). For instance, SAR image encompasses information regarding the shape, structure, and texture of ground objects [18], [19], [20]. The hyperspectral image provides abundant spatial–spectral information about ground objects [11], [12], [13], [14], [15], [16], [17]. Hence, the fusion of hyperspectral and SAR comprehensively utilizes the advantages of both to improve the accuracy and reliability of scene interpretation [28], [29].

To utilize the complementary information between multimodal remote sensing images, a significant quantity of work has been proposed. According to different feature extraction methods, these research tasks can be categorized into three distinct clusters: artificial features-based models, convolutional neural networks-based models, and attention networks-based models. The artificial features-based models usually utilize traditional mathematical and statistical models to extract features of multiple data. In [30], a multilevel structure extraction (MSE) method was proposed to integrate diverse remote sensing data. The MSE method constructs a multilevel structure to extract features by cascading morphological profiles and structural features. In [31], a shared and specific feature learning (S2FL) method was proposed. S2FL can decompose multisensor images into specific components to achieve multimodal information fusion. In [32], a subspace sensor fusion (SubFus) method was proposed to fuse heterogeneous data. The core idea of SubFus is to project multisource high-dimensional data into low-dimensional features. The above-mentioned methods can effectively improve the scene interpretation performance. However, these methods rely on subjectively designed features.

The convolutional neural networks-based models are to realize the fusion classification of multisource images by constructing an appropriate network architecture and cost function. In [33], a convolutional neural network (CNN) was introduced for the classification of multisource data fusion, in which one branch is dedicated to extracting features from HSIs, while the other branch focuses on extracting spatial features from LiDAR. Hong et al. [34] introduced the encoder–decoder network (EndNet) structure to achieve multimodal deep learning (MDL). In [35], coupled convolutional neural networks (Coupled CNNs) were implemented to merge multisource images, in which a parameter-sharing strategy is used to reduce the network parameters. Wang et al. [36] proposed a multisource data classification framework that pays

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more attention to the complementary multimodal spatial information and the specificity of HSI spectral information. In [37], a spatial–spectral cross-modal enhancement network (S^2ENet) was proposed. To facilitate information fusion between multisource images in the feature extraction stage, the S^2ENet designs a spatial–spectral fusion module. Compared with artificial features-based methods, deep learning models offer superior representation capabilities for multimodal remote sensing images.

The attention networks-based models often use the transformer architecture, which efficiently handles global dependencies and flexible inputs. This makes them ideal for capturing global context and processing long-range relationships in sequence data. In [38], a multimodal fusion transformer (MFT) was proposed. Unlike other fusion methods, other modal data in the MFT network are used to generate the external classification token of the transformer encoder to improve classification accuracy. In [39], an original deep hierarchical vision transformer (DHViT) structure was developed, in which the spectral transformer is applied to extract the spectral features of the HSI and the spatial transformer is utilized to extract the spatial features of the HSI and LiDAR. In [40], an extended visual transformer (ExViT) for multimodal remote sensing image classification is proposed. It designs a cross-modal attention module for feature fusion. In [41], a spatial–spectral perception network (S2PNet) was constructed, which emphasizes the advantages of different data sources and establishes spectral and spatial relationships respectively. It improves the classification accuracy of multisource remote sensing data. Zhao et al. [42] combined the strengths of CNNs and transformers by proposing a hierarchical CNN and transformer (HCT) model, overcoming the limitations of using a single architecture.

Although these methods can improve the classification performance, both transformers and CNNs-based methods still face two challenges. 1) The edge features in remote sensing images are often nonlinear and irregular. The CNNs-based methods cannot well represent the highly nonlinear structures. 2) Recent works often use attention mechanism-based modules for feature fusion, but they fail to leverage the abundant contextual information present between adjacent keys. Inspired by the advantage of graph convolutional network (GCN) in modeling the high-order nonlinear structures [43], [44], [45], [46], [47], [48], this paper article proposes a graph convolutional fusion network (GCFNet) for hyperspectral and SAR image classification. First, a spectral residual neural network is utilized to extract the spectral features of HSI. Then, a two-branch spatial GCN for extracting spatial features of multimodal remote sensing images is proposed. Finally, a cross-contextual transformer fusion module is designed to merge the extracted spectral and spatial features, and then the fused features are forwarded to the dense layer to yield the final prediction outcome. Experiments on several datasets verify the effectiveness of the GCFNet. To summarize, the key contributions of this task are as follows.

1) To better represent the nonlinear structures, a novel GCFNet is proposed for the joint classification of HSI and SAR, in which a spatial GCN is employed to mine the spatial features of multimodal remote sensing images.
2) To fully use the contextual information, a cross-context attention mechanism is designed to merge the heterogeneous features, which can mine the contextual information between adjacent keys of different modalities and effectively fuse the spatial and spectral features of multimodal remote sensing data.
3) Experiments on several datasets demonstrate that the proposed method consistently outperforms other multimodal remote sensing image fusion techniques in terms of subjective and objective results.

This article is organized as follows. Section II illustrates the precise procedure of GCFNet. Section III discusses the experimental results. Finally, Section IV presents the conclusions.

II. Proposed Method

In this section, a GCFNet is implemented for hyperspectral and SAR image classification. Fig. 1 presents the schematic of GCFNet, which is composed of three central parts: spectral convolutional neural network, spatial GCN, and cross-contextual transformer fusion module. Each module is detailed below.

A. Spectral Residual Neural Network

To accurately extract the spectral features of HSIs, a spectral residual CNN is constructed, which consists of a convolution layer for spectral transformation and three spectral residual modules for extracting spectral features.

1) Spectral Transformation: Hyperspectral data contain a lot of noise information. To improve the representation ability of the spectral convolutional network, a spectral transformation is utilized to process the input HSI data for dimension reduction ($\mathbf{X}_{\text{hsi}} \in R^{H\times W\times C}$, where H is the height, W denotes the width, and C represents the number of bands of the $\mathbf{X}_{\text{hsi}}$). Specifically, $\mathbf{X}_{\text{hsi}}$ is processed by a batch normalization layer, a 3×3 kernel 2-D CNN convolution layer, and a LeakyReLU activation layer to yield $\mathbf{X}_{\text{spe}} \in R^{H\times W\times C_1}$ ($C_1 = 32$ in the experiments).

2) Spectral Residual Module: After obtaining the dimension-reduced HSI, a spectral residual module is proposed for the extraction of spectral attributes. Fig. 2 displays the flowchart of the spectral residual module. Specifically, two 1×1 convolutional layers are utilized to extract spectral attributes, and then, a residual skip connection is applied to enhance the effectiveness of the model. In addition, to speed up the training of the network, a batch normalization operation is performed before convolution and the activation function LeakyReLU after convolution is introduced to enhance the nonlinearity ability of the model. The spectral residual calculation can be expressed as

$$\widehat{\mathbf{X}}_0 = \mathbf{X}_0 + F(F(\mathbf{X}_0)) \tag{1}$$

$$F(\mathbf{X}_0) = \text{ReLu}(\text{Conv}(\text{Bn}(\mathbf{X}_0))) \tag{2}$$

where $\mathbf{X}_0$ represents the input data, $\widehat{\mathbf{X}}_0$ represents the output data, Bn($\bullet$) is the batch normalization operation, ReLu($\bullet$)

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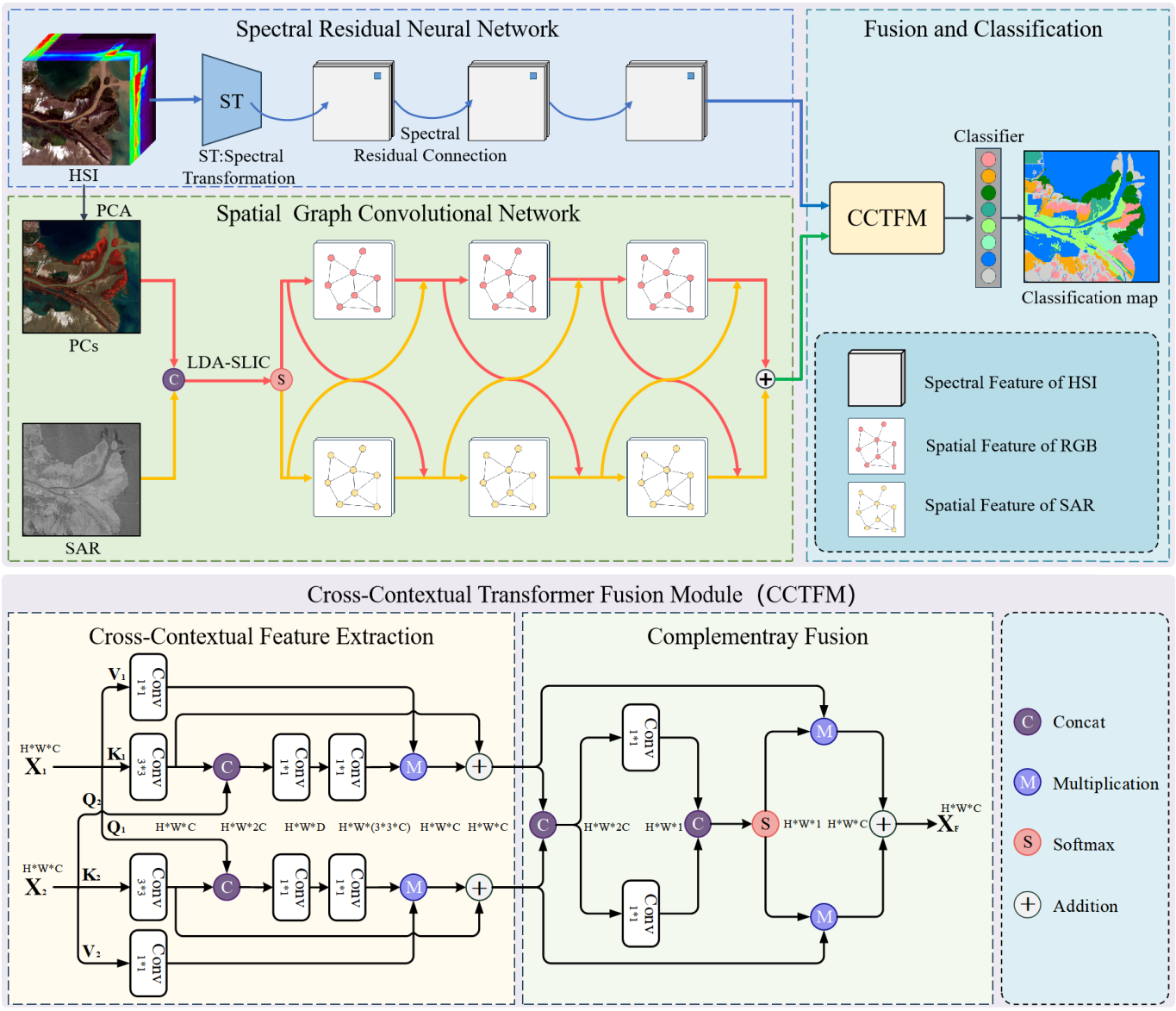


Fig. 1. Schematic illustration of GCFNet. The matrix dimension D in the figure is the greatest integer less than or equal to $C/2$.

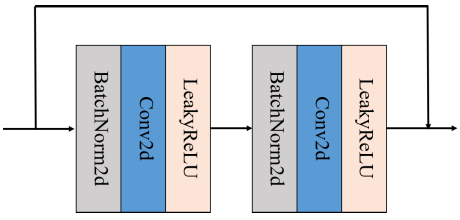


Fig. 2. Structure of the spectral residual connection.

denotes the activation function, and Conv(•) represents the convolution with its convolution kernel, step size, and padding are 1, 1, and 0, respectively.

B. Spatial GCN

Given the wide range of ground objects and their intricate nature in remote sensing images, traditional convolutional neural networks cannot well represent the complicated spectral–spatial structures since they only consider the fixed-window information while ignoring the correlation among neighboring regions. To this end, a dual-branch GCN is proposed to extract the spatial features of multimodal images. The network consists of a superpixel segmentation module and two GCN branches with the same structure. Furthermore, to integrate the spatial features of diverse modalities, the two branches are connected through cross-residual skip connections.

1) Superpixel Segmentation Module: When each pixel of the input data is viewed as a graph node in GCN, a huge computational cost will be incurred. To address this issue, a commonly employed approach is to model GCN in the superpixel level instead of the pixel level [45], [46]. Therefore, a superpixel segmentation operation is adopted before feature extraction. Specifically, we utilize linear discriminant analysis (LDA) for dimensionality reduction of the input image. Then, simple linear iterative clustering (SLIC) is used to divide

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the data into many superpixel blocks. This process can be expressed as

$$\mathbf{S} = \mathrm{LDA} - \mathrm{SLIC}\left(\mathbf{X}_{\mathrm{spa}}\right) = \{\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_k\} \tag{3}$$

$$\mathbf{X}_{\mathrm{spa}} = \left[\mathbf{X}_{\mathrm{pcs}}, \mathbf{X}_{\mathrm{sar}}\right] = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{HW}\} \tag{4}$$

where $\mathbf{X}_{\mathrm{sar}}$ is SAR data, $\mathbf{X}_{\mathrm{pcs}} \in R^{H\times W\times 3}$ is the pseudo-color data, which is obtained by performing PCA on HSI data $\mathbf{X}_{\mathrm{hsi}}$, $\mathbf{X}_{\mathrm{spa}}$ is the spatial feature data to be processed, $\mathbf{x}_i$ is the ith pixel in $\mathbf{X}_{\mathrm{spa}}$, $\mathbf{S}$ is the set of superpixel blocks, $\mathbf{s}_i$ is the ith superpixel block in $\mathbf{S}$. $k = H \times W/\lambda$, and the parameter λ regulates the dimensions of superpixels. Furthermore, the assignment matrix $\mathbf{M} \in R^{HW\times k}$ for pixel to superpixel mapping can be obtained by

$$\mathbf{M}_{ij} = \begin{cases} 1, & \text{if } \mathbf{x}_i \in \mathbf{s}_i \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where $\mathbf{M}_{ij}$ represents the value of $\mathbf{M}$ at location (i, j). At the same time, to ensure simplicity and avoid bias, constructing an unweighted graph with an adjacency mask that depicts the connectivity between nodes is crucial. This is achieved by defining the adjacency mask $\mathbf{A}$ as a matrix of size $k \times k$, where k denotes the count of nodes in the graph. The adjacency mask $\mathbf{A}$ can be expressed as follows:

$$\mathbf{A}_{ij} = \exp\left(-\left\|\mathbf{s}_i - \mathbf{s}_j\right\|^2\right) \tag{6}$$

where $\mathbf{A}_{ij}$ represents the value of $\mathbf{A}$ at location (i, j).

2) Two-Branch GCN: After performing superpixel segmentation, the constructed graph structure data consist of the node matrix $\mathbf{H}$ and the affinity matrix $\mathbf{A}$. Accordingly, the convolution operation of GCN is shown as

$$\mathbf{H}^{(n+1)} = S\left(\widehat{\mathbf{D}}^{-\frac{n}{2}}\widehat{\mathbf{A}}\widehat{\mathbf{D}}^{-\frac{n}{2}}\mathbf{H}^{(n)}\mathbf{W}^{(n+1)}\right) \tag{7}$$

$$\widehat{\mathbf{A}} = \mathbf{A} + \mathbf{I}, \quad \widehat{\mathbf{D}}_{ij} = \sum_j \widehat{\mathbf{A}}_{ij} \tag{8}$$

where $\mathbf{W}^{(n)}$ is the trainable parameter, $\mathbf{A}$ represents the adjacency matrix, $\mathbf{D}$ denotes the degree matrix, $\mathbf{H}^{(n)}$ is the output of the nth layer, $S(\bullet)$ is the activation function. Then, (7) and (8) can be reformulated as a function, that is,

$$\mathbf{H}^{(n+1)} = g\left(\mathbf{H}^{(n)}, \mathbf{A}\right) \tag{9}$$

where $g(\bullet)$ is a GCN operation. To better fuse the complementary information between the input features, a cross-modal residual connection method is utilized, which is shown as follows:

$$\mathbf{H}_{\mathrm{pcs}}^{(n+1)} = g\left(\mathbf{H}_{\mathrm{pcs}}^{(n)}, A\right) + \mathbf{H}_{\mathrm{sar}}^{(n)} \tag{10}$$

$$\mathbf{H}_{\mathrm{sar}}^{(n+1)} = g\left(\mathbf{H}_{\mathrm{sar}}^{(n)}, A\right) + \mathbf{H}_{\mathrm{pcs}}^{(n)} \tag{11}$$

where $\mathbf{H}_{\mathrm{sar}}^{(n)}$ and $\mathbf{H}_{\mathrm{sar}}^{(n)}$ are the output of the nth layer SAR and PCs data, respectively. The input features of the first layer are $[\mathbf{H}_{\mathrm{pcs}}^{(0)}, \mathbf{H}_{\mathrm{sar}}^{(0)}] = \mathbf{H}$.

C. Cross-Contextual Transformer Fusion Module

After extracting the spectral features and spatial features of multimodal data, how to fuse the features to enhance the accuracy of ground object classification is still an important issue. To this end, this article designs a cross-contextual transformer-based fusion module, which consists of cross-context feature extraction and feature fusion.

1) Cross-Contextual Feature Extraction: Traditional transformers have not fully utilized the rich context between adjacent keys [49]. To alleviate this issue, a contextual transformer is exploited to extract the cross-contextual feature. In more detail, assume $\mathbf{X}_1$ be the input feature, the cross-contextual features $\widehat{\mathbf{X}}_1$ can be calculated as follows:

$$\mathbf{Q}_2 = \mathbf{X}_2, \mathbf{K}_1 = \mathrm{Conv}(\mathbf{X}_1, \mathbf{P}_K), \quad \mathbf{V}_1 = \mathbf{X}_1\mathbf{P}_V \tag{12}$$

where $\mathbf{X}_1$ is the input feature, the $\mathbf{V}_1$ is obtained through a convolution operation with a convolution kernel of 1×1, while $\mathbf{K}_1$ is obtained through a convolution operation with a convolution kernel of 3×3. The parameters $\mathbf{P}_V$ and $\mathbf{P}_K$ are trainable. Next, the attention matrix $\mathbf{A}_t$ including contextual information is calculated as follows:

$$\mathbf{A}_t = [\mathbf{K}_1, \mathbf{Q}_2]\mathbf{P}_1\mathbf{P}_2 \tag{13}$$

where $\mathbf{P}_1$ and $\mathbf{P}_2$ are trainable parameters. Finally, the output $\widehat{\mathbf{X}}_1$ is calculated by

$$\widehat{\mathbf{X}}_1 = \mathbf{V}_1 \times \mathbf{A}_t + \mathbf{K}_1 \tag{14}$$

where $\widehat{\mathbf{X}}_1$ is the output feature. Similarly, the cross-contextual features $\widehat{\mathbf{X}}_2$ can be obtained.

2) Complementray Fusion: To fuse the cross-contextual features $\widehat{\mathbf{X}}_1$ and $\widehat{\mathbf{X}}_2$, a weighted fusion method is designed. First, the extracted cross-contextual features are connected, and then, two parallel convolution operations are used to obtain the initial weights

$$\mathbf{G}_1 = \mathrm{Conv}_1\left(\left[\widehat{\mathbf{X}}_1, \widehat{\mathbf{X}}_2\right]\right) \tag{15}$$

$$\mathbf{G}_2 = \mathrm{Conv}_2\left(\left[\widehat{\mathbf{X}}_1, \widehat{\mathbf{X}}_2\right]\right) \tag{16}$$

where $\mathbf{G}_1$ and $\mathbf{G}_2$ are initial weights, $\mathrm{Conv}_1(\bullet)$ and $\mathrm{Conv}_2(\bullet)$ represent a convolution operation with 1×1 kernel. Next, the initial weights are mapped into [0, 1] as follows:

$$[\mathbf{A}_1, \mathbf{A}_2] = \mathrm{Softmax}([\mathbf{G}_1, \mathbf{G}_2]) \tag{17}$$

where $\mathbf{A}_1$ and $\mathbf{A}_2$ are normalized weights. Finally, the weighted fusion rule is adopted to merge $\widehat{\mathbf{X}}_1$ and $\widehat{\mathbf{X}}_2$ to obtain the final features

$$\mathbf{X}_F = \widehat{\mathbf{X}}_1 \times \mathbf{G}_1 + \widehat{\mathbf{X}}_2 \times \mathbf{G}_2 \tag{18}$$

where $\mathbf{X}_F$ is the fused features that are forwarded to the dense layer to yield the final prediction outcome.

III. Experiments

A. Datasets and Quality Assessment

To assess the effectiveness of the GCFNet, three datasets are utilized. The visualization results of the three datasets are shown in Figs. 3–5. Each dataset includes an HSI and a SAR image. Detailed descriptions of the three datasets will be given in Sections III-A1–III-A3.

1) Berlin Dataset: The dataset was collected in the city of Berlin and includes an HSI and a SAR. The HSI has a spatial

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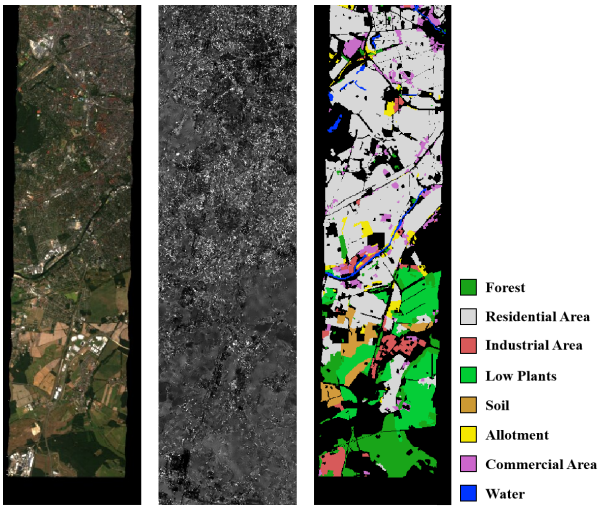


Fig. 3. Berlin dataset. (a) HSI false color composite. (b) SAR image. (c) Ground truth image.

dimension of 797 × 220 pixels, with a spatial resolution of 30 m. The spectrum range of this image spans from 0.40 to 2.50 μm, and it is equipped with 244 spectral channels. The SAR image obtained from the Sentinel-1 satellite encompasses the identical region and possesses a spatial dimension of 1723 × 476 pixels, with a spatial resolution of 13.89 m. This image comprises four polarization images. To achieve uniform spatial dimensions, the HSI image is resampled to match the spatial scale of the SAR image [31].

2) Augsburg Dataset: The second dataset is the Augsburg dataset, comprising hyperspectral and SAR images captured in Augsburg, Germany. The data are categorized into seven distinct groups based on the various sorts of ground objects. The HySpex sensor collects hyperspectral imagery with a spatial dimension of 332 × 485 pixels, providing a spatial resolution of 30 m. The image is composed of 180 spectral bands that span a range of wavelengths from 0.40 to 2.50 μm. The SAR image is obtained using the Sentinel-1 sensor and possesses identical spatial dimensions and resolution as the HSI. SAR data have four features according to polarization decomposition [50].

3) YR Dataset: The third dataset is the YR dataset, which comprises eight categories of representative objects found in the Yellow River Delta. The advanced hyperspectral imager (AHSI) onboard the Gaofen-5 satellite collects HSI over the Yellow River Delta. The HSI data are subjected to various processing stages, such as atmospheric correction, geometric correction, removal of invalid bands, and image cropping. The dataset consists of 330 spectral bands, with center wavelengths ranging from 0.39 to 2.51 μm, with a spatial resolution of 30 m. After removing bands with a low signal-to-noise ratio, a total of 296 spectral bands are kept for experimental study. The data collected by the Sentinel-1A sensor are SAR data [51], [52].

4) Quality Assessment: To objectively assess the classification effectiveness of the GCFNet model, three evaluation metrics, namely overall accuracy (OA), average accuracy (AA), and the Kappa coefficient, are utilized. These indicators provide quantitative measures of the accuracy and agreement between the predicted outcome and the reference data.

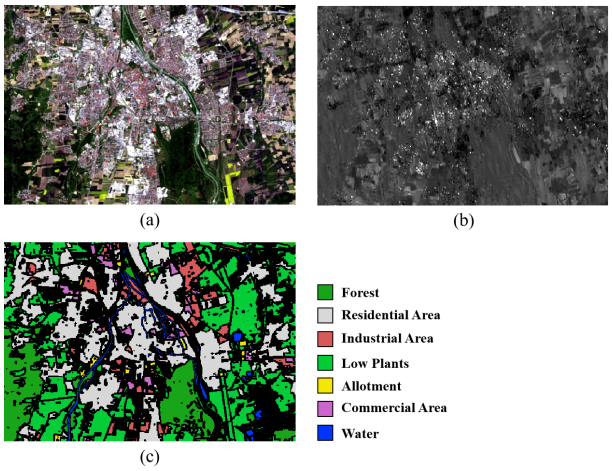


Fig. 4. Augsburg dataset. (a) HSI false color composite. (b) SAR image. (c) Ground truth image.

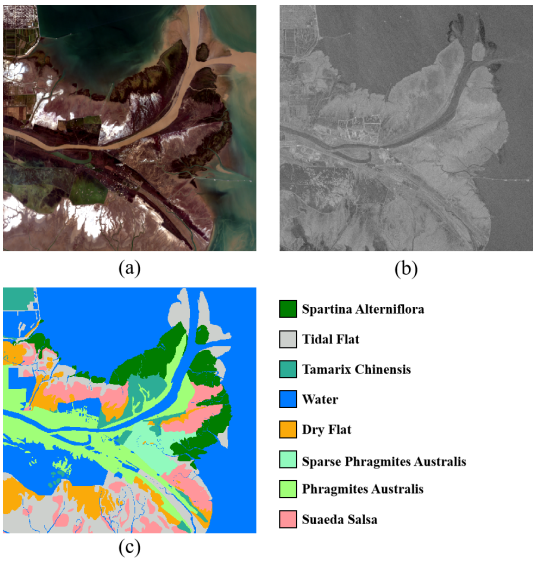


Fig. 5. YR dataset. (a) HSI false color composite. (b) SAR image. (c) Ground-truth image.

B. Experimental Setup

To verify the performance of the GCFNet model, several representative methods are used for comparison.

1) *MSE [30]:* The MSE method is a multilevel structure extraction-based multimodal image fusion method, which mainly contains multilevel structure feature extraction and low-rank model-based feature fusion.
2) *S2FL [31]:* The S2FL method is a multisource image classification model based on a shared specific feature learning model, in which the multisource remote sensing images are decomposed into specific components to achieve more efficient mixing of multimodal information.
3) *Multimodal Deep Learning [53]:* The MDL is a multimodal deep learning architecture for multisource image classification. In this work, the cross-fusion strategy is adopted.

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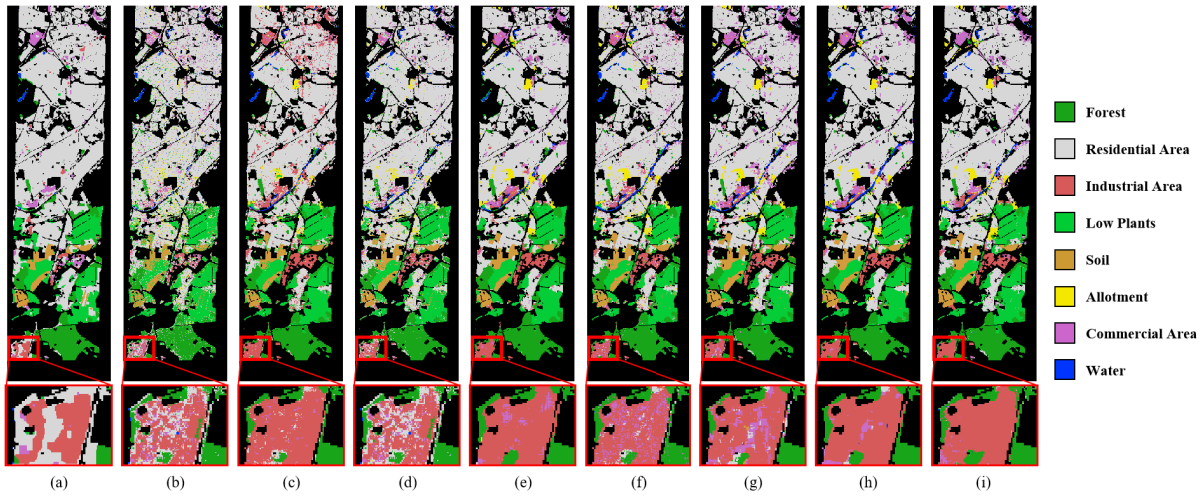


Fig. 6. Visualization results and local magnification of all fusion methods on the Berlin dataset. (a) MSE. (b) S2FL. (c) MDL. (d) EndNet. (e) CALC. (f) DHViT. (g) MFT. (h) ExViT. (i) GCFNet.

4) *EndNet [34]:* The EndNet is an encoder–decoder network framework for multisensor data joint classification, in which a reconstruction strategy is used to activate neurons across modalities.
5) *CALC [54]:* The CALC is a classification method based on coupled adversarial learning for joint classification of HSI and LiDAR. Its optimization function during the training stage combines contrastive learning loss and classification loss.
6) *DHViT [39]:* The DHViT is a deep transformer network for multimodal image classification. The long-range dependency modeling is used to fuse and classify multisource remote sensing images.
7) *MFT [38]:* The MFT is a transformer network used for multimodal fusion classification of remote sensing data. It uses the vision transformer model for HSI fusion classification tasks.
8) *ExViT [40]:* The ExViT is an extended visual transformer model for multimodal remote sensing image classification, which uses a cross-modal attention module to fuse heterogeneous features.
9) *CEGCN [46]:* The CEGCN is a CNN-enhanced GCN model for HSI classification. It combines the advantages of CNN and GCN to model at the pixel level and superpixel level, respectively.

The proposed method is implemented with Python language programming on the PyTorch framework. The segmentation scales (λ) of the Berlin, Augsburg, and YR datasets are set to 500, 100, and 200, respectively. The corresponding initial superpixel numbers are 2020, 1580, and 3045, respectively. The network is trained with the Adam optimization algorithm. The training process consists of 1000 epochs, with a learning rate set to 0.005. For all comparison models, the parameter settings refer to the original publications.

C. Classification Performance

In this section, the classification performance of all considered models is evaluated, in which the number of training sets is 1%. To more intuitively compare the classification effectiveness of all considered models, the visualization results of all methods on the three datasets are compared in detail.

1) Experiment on the Berlin Dataset: The classification visualization results of all fusion models on the Berlin dataset are illustrated in Fig. 6. It is found that there are a multitude of misclassified categories in the classification maps of MSE and S2FL. In particular, Allotment is not recognized correctly. The CNNs-based classification methods have better classification performance. However, in the classification map, a multitude of commercial areas are incorrectly classified as residential areas or industrial areas. In contrast, the classification effect of Fig. 6(e)–(g) is further improved. In particular, the proposed GCFNet method can correctly identify residential areas and commercial areas. In the enlarged area, the proposed method has the least misclassified points.

It can be seen from the objective indicator results in Table I that two traditional machine learning methods, MSE and S2FL, have extremely low classification accuracy in some categories, and OA is also the lowest among all methods. The objective indicators obtained by the CNNs-based methods are improved, and the OA has increased by at least 3.13%. The objective results of the Transformer network are further improved. Overall, the proposed GCFNet method obtains the highest objective metric results. Specifically, compared with other competitors, GCFNet gains OA improvements of 9.99%, 12.79%, 6.78%, 6.86%, 1.30%, 4.68%, 2.73%, 1.18%, and 2.89%, respectively.

2) Experiment on the Augsburg Dataset: The visualization results of all fusion models on the Augsburg dataset are shown in Fig. 7. From the classification maps, it can be seen that the MSE yields the worst results, and other categories except water fail to achieve ideal results. The S2FL method classifies a multitude of residential areas into industrial areas. The classification visualization effect of the deep learning methods has been slightly improved. However, the three methods, i.e., MDL, EndNet, CALC, DHViT, and ExViT, still fail to classify commercial areas correctly. The MFT can roughly identify commercial areas, but there are still parts in the middle that are mistakenly classified as industrial areas. By contrast,

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TABLE I

OBJECTIVE CLASSIFICATION INDICATORS OBTAINED BY ALL CONSIDERED MODELS ON THE BERLIN DATASET

Class No.	Artificial features-based models		CNNs-based models			Attention networks-based models			GCNs-based models	
	MSE	S2FL	MDL	EndNet	CALC	DHViT	MFT	ExViT	CEGCN	GCFNet
1	0.8687	0.7901	0.8721	0.8291	**0.9229**	0.8745	0.8529	0.8968	0.8803	0.8635
2	0.8095	0.8261	0.9022	0.9380	0.9481	0.9151	0.9254	0.9509	0.9677	**0.9702**
3	0.6217	0.5005	**0.8201**	0.4429	0.7654	0.6321	0.6384	0.6950	0.6928	0.7826
4	0.7755	0.8097	0.9401	0.8591	0.9380	0.9175	0.9265	0.8198	0.9326	**0.9492**
5	0.8711	0.7776	0.7737	0.8624	0.8144	0.8739	0.9020	0.8991	**0.9330**	0.9144
6	0.0000	0.2644	0.3926	0.3878	0.6231	0.6473	**0.6788**	0.6279	0.6339	0.5734
7	0.5807	0.3328	0.1125	0.2292	0.4820	0.3004	**0.5899**	0.4635	0.3264	0.4585
8	**0.9523**	0.6556	0.6357	0.6384	0.4300	0.8488	0.6646	0.8281	0.7862	0.7102
OA	0.8032	0.7752	0.8353	0.8345	0.8901	0.8563	0.8758	0.8913	0.8742	**0.9031**
AA	0.6849	0.6196	0.6811	0.6483	0.7405	0.7512	0.7723	0.7726	0.7691	**0.7778**
Kappa	0.6537	0.6267	0.7387	0.7218	0.8223	0.7701	0.8025	0.8244	0.8154	**0.8401**

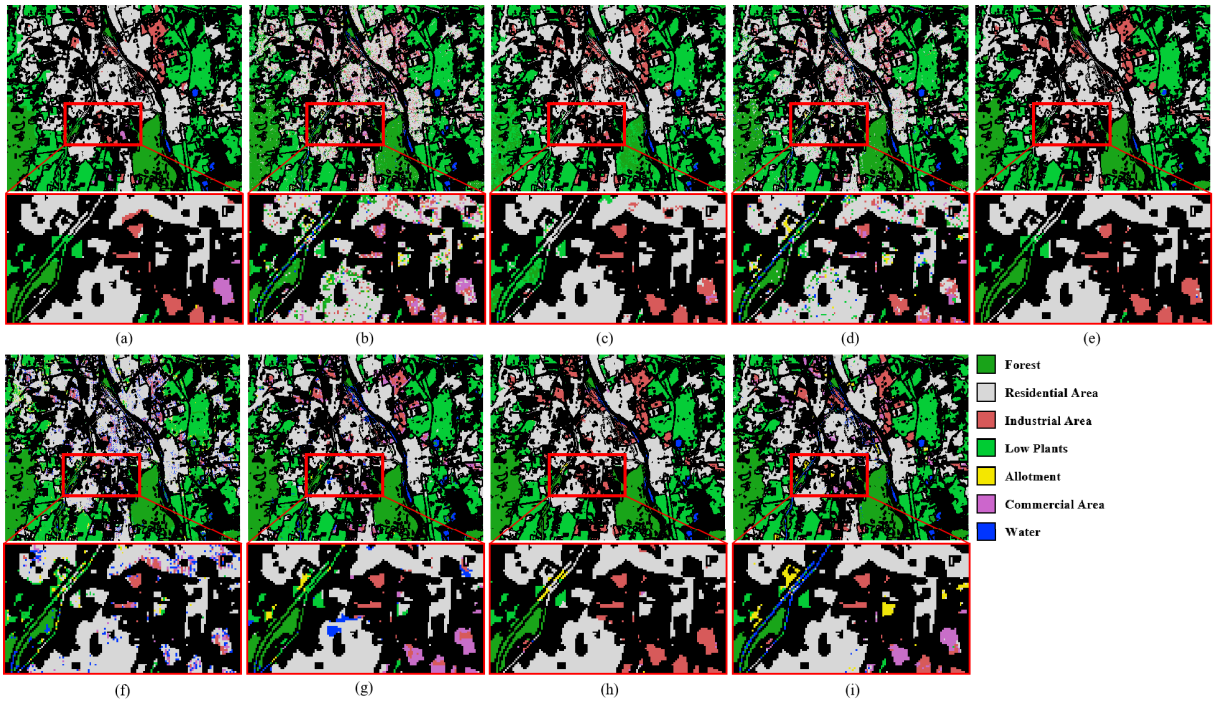


Fig. 7. Visualization results and local magnification of all fusion methods on the Augsburg dataset. (a) MSE. (b) S2FL. (c) MDL. (d) EndNet. (e) CALC. (f) DHViT. (g) MFT. (h) ExViT. (i) GCFNet.

TABLE II

OBJECTIVE CLASSIFICATION INDICATORS OBTAINED BY ALL CONSIDERED MODELS ON THE AUGSBURG DATASET

Class No.	Artificial features-based models		CNNs-based models			Attention networks-based models			GCNs-based models	
	MSE	S2FL	MDL	EndNet	CALC	DHViT	MFT	ExViT	CEGCN	GCFNet
1	0.8234	0.8330	0.4643	0.9188	0.9112	0.9091	0.9585	0.9538	**0.9778**	0.9606
2	0.8646	0.8591	**0.9839**	0.8914	0.9796	0.8784	0.9643	0.9742	0.9677	0.9696
3	0.5496	0.3916	0.7718	0.3932	0.7762	0.4937	0.5136	**0.8843**	0.7120	0.6311
4	0.8449	0.9391	**0.9734**	0.9497	0.9361	0.9341	0.9267	0.9452	0.9460	0.9728
5	0.0000	0.2099	0.0019	0.1797	0.0000	0.7224	0.7491	0.0096	0.1434	**0.7631**
6	0.2933	0.2020	0.0183	0.1306	0.0024	0.4325	0.4835	0.0263	0.0262	**0.6513**
7	**0.7586**	0.7057	0.3258	0.5116	0.1666	0.3382	0.4486	0.3995	0.0059	0.5227
OA	0.8305	0.8356	0.8405	0.8633	0.8998	0.8751	0.9025	0.9086	0.9012	**0.9197**
AA	0.5906	0.5915	0.5056	0.5679	0.5389	0.6726	0.7206	0.5990	0.5399	**0.7816**
Kappa	0.7555	0.7659	0.7620	0.8022	0.8554	0.8353	0.8602	0.8630	0.8615	**0.8694**

the classification result of the GCFNet model is the best, and the commercial area is basically classified correctly. On the local magnification graph, GCFNet has the best classification results for allotment and industrial areas.

Table II gives the objective metric results of each classification model on the Augsburg dataset. The GCFNet model obtains the highest classification accuracy on most ground objects and has also the best results in OA, AA, and Kappa objective indicators. First, compared with

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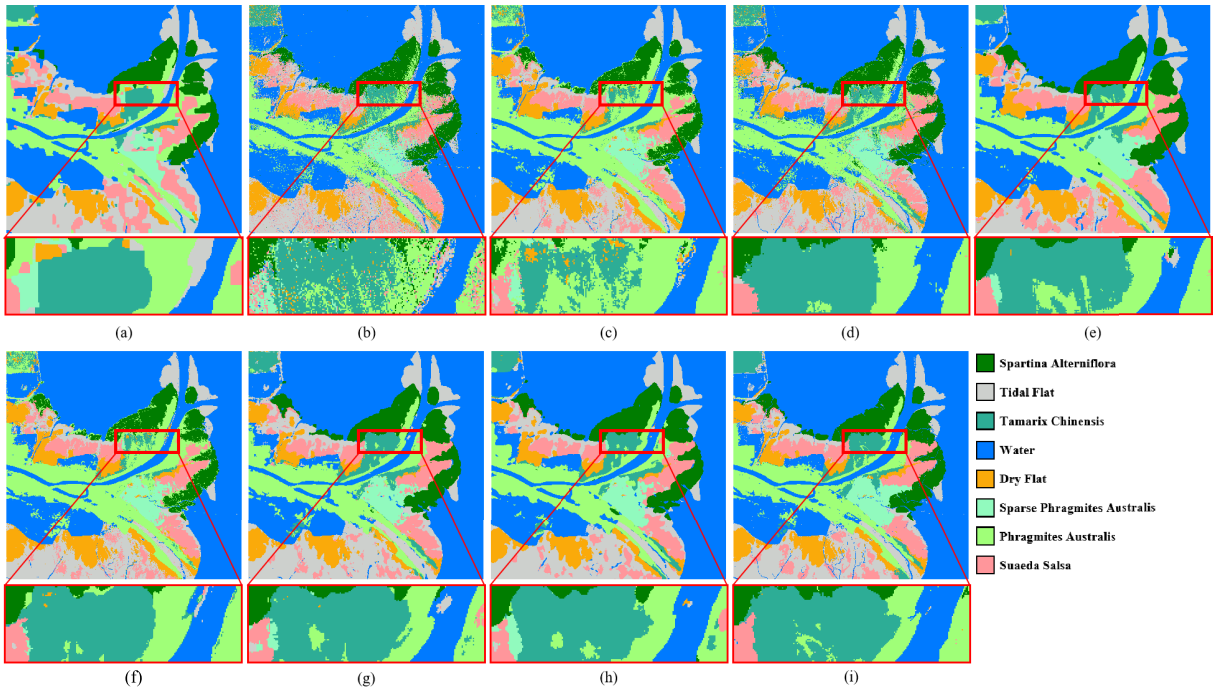


Fig. 8. Visualization results and local magnification of all fusion methods on the YR dataset. (a) MSE. (b) S2FL. (c) MDL. (d) EndNet. (e) CALC. (f) DHViT. (g) MFT. (h) ExViT. (i) GCFNet.

TABLE III
OBJECTIVE CLASSIFICATION INDICATORS OBTAINED BY ALL CONSIDERED MODELS ON THE YR DATASET

Class No.	Artificial features-based models		CNNs-based models			Attention networks-based models			GCNs-based models	
	MSE	S2FL	MDL	EndNet	CALC	DHViT	MFT	ExViT	CEGCN	GCFNet
1	0.7748	0.8172	**0.9725**	0.8754	0.9326	0.8839	0.9048	0.8845	0.9038	0.9261
2	0.9426	0.8968	0.8094	0.9034	**0.9575**	0.8965	0.9418	0.9308	0.9096	0.9162
3	0.9398	0.9526	0.9539	0.9653	0.9436	0.9494	0.9506	0.9616	0.9617	**0.9664**
4	0.6985	0.7100	0.3727	0.6785	0.7757	0.8008	**0.8586**	0.8220	0.8255	0.8424
5	0.7647	0.7398	0.7185	0.7971	0.9167	0.8636	0.8268	0.8048	0.8990	**0.9348**
6	0.6093	0.5917	0.7362	0.7365	**0.8822**	0.7619	0.7899	0.7540	0.8033	0.8322
7	0.6953	0.7288	0.8010	0.8025	0.7033	0.8614	**0.9149**	0.8738	0.8664	0.8702
8	0.8829	0.8408	**0.9426**	0.8032	0.8999	0.8152	0.8238	0.8636	0.8291	0.8549
OA	0.8384	0.8475	0.8748	0.8821	0.8949	0.8973	0.9113	0.9050	0.9091	**0.9205**
AA	0.7885	0.7847	0.7884	0.8202	0.8764	0.8541	0.8764	0.8619	0.8748	**0.8929**
Kappa	0.7818	0.7952	0.8321	0.8414	0.8603	0.8487	0.8814	0.8724	0.8781	**0.8933**

MSE and S2FL methods, GCFNet improves OA by 8.94% and 8.41%, respectively. Second, compared with the three methods based on convolutional neural networks, the GCFNet method also improves the classification accuracy by at least 1.99%. Finally, even compared with transformer networks, i.e., DHViT, MFT, and ExViT, the three quantitative evaluation indicators of the GCFNet method are still improved.

3) Experiment on the YR Dataset: The visualization results of all fusion classification models on the YR dataset are depicted in Fig. 8. First, the two traditional machine learning classification methods perform poorly. Many edges in the visualization results of the MSE method are not classified correctly, while the classification result map of the S2FL model contains a multitude of noise points. Next, for the CNNs and Transformer-based methods, the effect of the visualization result has been improved to a certain extent, but it still contains a small amount of noise. For example, Tamarix Chinensis is mistakenly divided into Phragmites Australis at different scales. Finally, the proposed GCFNet method identifies the Tamarix Chinensis class and achieves superior classification outcomes. It can also be seen from the local enlarged image that the proposed model is basically correct in classifying Tamarix Chinensis, and the number of misclassification points for water is also the least.

The objective indicators of each model are presented in Table III. As observed from the table, the MSE has the best classification accuracy on Tidal Flat, but the classification accuracies on other ground objects are not ideal. In comparison, the OA of the S2FL has been improved. The classification accuracies of the three convolutional neural network methods are further improved. The Transformer network yields relatively good results. Among them, MFT attains the highest classification accuracies in Water and Phragmites Australis. Finally, the GCFNet achieves the best classification accuracies for most of the ground objects, and the quantitative evaluation indicators, i.e., OA, AA, and Kappa are also the highest among all compared models.

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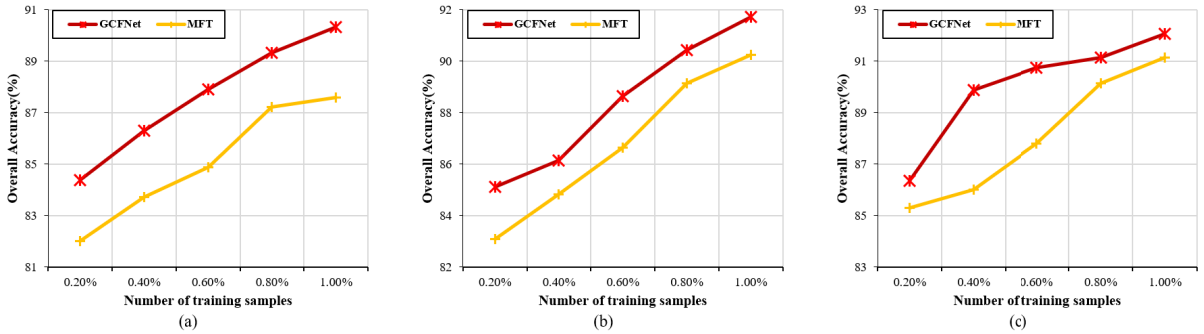


Fig. 9. OA with different sizes of training samples. (a) Berlin dataset. (b) Augsburg dataset. (c) YR dataset.

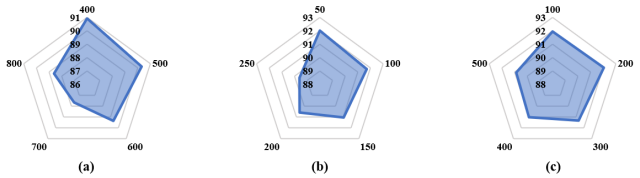


Fig. 10. OA of different superpixel segmentation scales (λ) on three datasets. (a) Berlin dataset. (b) Augsburg dataset. (c) YR dataset.

D. Ablation Studies

1) Influence of Various Training Sample Sizes: In this section, the influence of various training sample sizes is analyzed. To better illustrate the effect of the GCFNet, the MFT is applied as the competitor since it yields the best classification results among all compared techniques. Fig. 9 illustrates the quantitative outcomes of the three datasets. The training set size ranges from 0.2% to 1% with increments of 0.2%. It is found that the OA of the two models increases when the size of training samples is expanded. Moreover, the GCFNet model yields better classification results, which also further demonstrates the effectiveness of the GCFNet.

2) Influence of Superpixel Segmentation Scales: The choice of superpixel segmentation scale (λ) plays a critical role in the fusion classification performance. Fig. 10 shows the OA of the proposed method with different superpixel segmentation scales on three datasets. A much larger λ easily leads under segmentation. Moreover, too much smaller λ may generate that each superpixel cannot contain enough spatial information. As shown in Fig. 10, it can be seen that the classification performance obtained by the proposed method on three datasets tends to decrease when the λ value increases. The reason is that each superpixel contains different types of land covers. When the λ value is relatively large, the computing time of the proposed method tends to increase. To balance classification accuracy and computational efficiency, λ values for the Berlin, Augsburg, and YR datasets are set to 500, 100, and 200, respectively.

3) Effect of the Proposed Fusion Framework: To demonstrate the performance of the proposed fusion framework and modules, this section includes a series of ablation studies. The results of the ablation experiments are recorded in Table IV. First, for the ablation experiments of the GCFNet framework, the OA of the GCFNet is significantly improved compared with the three independent modes. Considering the Berlin dataset as a case in point, the OA of GCFNet is 6.06%, 4.34%, and 7.03% higher than the HSI, SAR, and RAB modes, respectively. Secondly, if the spatial feature cross-fusion method CMRes is not used, the accuracy on the three datasets decreases by 1.41%, 2.63%, and 1.49%, respectively. Finally, through the comparative experiments of the fusion module CCTFM, it was found that the fusion of spatial and spectral features can indeed improve the discriminative ability of the model. The OA of the three datasets increased by 1.96%, 1.67%, and 1.74%, respectively.

TABLE IV
ABLATION EXPERIMENTS ON THE GCFNet ON THREE DATASETS

Modalities			Fusion Module		Overall Accuracy (%)		
HSI	SAR	PCs	CMRes	CCTFM	Berlin	Augsburg	YR
✓					84.25	89.23	88.67
	✓				85.94	86.18	83.17
		✓			83.28	88.16	87.41
✓	✓				87.10	89.41	89.47
✓		✓			86.26	89.37	88.77
	✓	✓			86.32	88.63	87.93
✓	✓	✓			87.57	89.57	89.68
✓	✓	✓		✓	88.89	89.88	90.56
✓	✓	✓	✓		88.35	90.04	90.31
✓	✓	✓	✓	✓	**90.31**	**91.71**	**92.05**

TABLE V
RUNNING TIME OF ALL FUSION METHODS ON THREE DATASETS

Methods	Stage	Running time (s)		
		Berlin	Augsburg	YR
MSE	Train	608.45	117.34	490.53
	Test	4.38	0.42	2.84
S2FL	Train	89.86	6.06	47.44
	Test	15.23	0.49	16.09
MDL	Train	451.95	67.02	560.89
	Test	95.27	13.98	63.15
EndNet	Train	279.45	49.16	341.38
	Test	29.74	5.32	20.32
CALC	Train	459.21	65.97	737.68
	Test	519.88	28.23	366.59
DHViT	Train	77823.15	7782.12	52784.95
	Test	6638.48	1063.84	4938.16
MFT	Train	823.41	78.97	949.57
	Test	251.27	15.37	168.27
ExViT	Train	570.72	90.71	904.37
	Test	371.65	36.49	245.29
GCFNet	Train	193.79	34.99	172.10
	Test	19.27	3.49	15.82

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4) Running Time: In this section, the computational cost of all models is discussed. All codes were executed on a desktop computer with a CPU of 3.70 GHz, memory of 64.0 GB, and a GPU of RTX 3090. The traditional machine learning models are developed in MATLAB, and all deep learning models are coded in Python language based on the PyTorch framework. The cost of calculation of all fusion models on the three datasets is recorded in Table V. Since high-dimensional feature mapping and learning are not required, traditional machine learning models, especially S2FL, take less total time to train and test. For deep learning methods, due to the structure complexity of the designed networks, they usually require more computing time. By contrast, the proposed GCFNet model takes the least time among all deep learning methods.

IV. Conclusion

In this article, we propose a GCFNet for integrating HSI and SAR images. First, a CNN extracts spectral features from the HSI data. Then, a dual-branch GCN captures the spatial features from both HSI and SAR images. To combine these features, we introduce a CCTFM, which merges spectral and spatial information. The proposed approach leverages the strengths of CNN for spectral modeling and GCN for spatial feature extraction. Extensive experiments on three benchmark datasets demonstrate that GCFNet outperforms several state-of-the-art techniques in classification accuracy. Additionally, GCFNet achieves the fastest computation time compared to other deep learning models. In the future, we will further explore the effectiveness of the proposed GCFNet for other data fusion.

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